Multirotor Flight Testing: Clarification for DARPA LIFT Challenge

How Test Flights Support Our Primary UAS Design

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Date	June 11, 2026
Re	DARPA LIFT Challenge — Response to Reviewer Query

Executive Summary

The multirotor flight tests visible in our uploaded footage are physics validation and calibration flights, not demonstrations of our final competition design. These flights are a deliberate step in our AI-driven design pipeline: we use real flight data to ground-truth the physics models that our optimizer relies on before committing designs to hardware. Our primary UAS design concept is detailed in the attached Concept Paper (Anote_Final_Concept_Paper.pdf).

How Multirotor Testing Fits the Design Pipeline

Stage 1 — Physics Model Calibration (Multirotor Test Flights)

We fly off-the-shelf multirotor platforms (quadcopters and hexacopters) to validate the core physics relationships our optimizer uses. The test vehicles are not the competition design — they are instrumented benches used to confirm that our simplified actuator disk model, structural equations, and feasibility constraints reflect real-world behavior before we run the optimizer at scale.

Model Parameter	Formula	Validated Against
Thrust	$T = 10 \times N_{motors} \times D_{prop}^2 \times throttle$	Hover thrust at varying throttle
Payload capacity	$payload = (T / 9.81) - total\_mass$	Known payload added incrementally
Structural safety factor	$mean\_UTS / 100$ MPa	Load deflection of CF vs. Al frames
Thrust-to-weight ratio	$T / (mass \times 9.81) \geq 1.5$	Minimum throttle at various mass configs

Stage 2 — AI-Driven Design Optimization

With validated physics models, we run our optimization pipeline (src/darpalyft/) over a large design space. The optimizer maximizes score = payload_kg / total_mass_kg subject to all physics and structural constraints, using Pareto front analysis to surface non-dominated designs.

- Materials: carbon fiber, aluminum alloy, titanium, fiberglass, ABS plastic
- Motor count: 4, 6, or 8
- Propeller diameter: 0.2 – 0.5 m
- Battery capacity: 400 – 1,200 Wh
- Component masses: per-component continuous variables

Stage 3 — Primary Design Selection & Refinement

The highest-scoring feasible designs from Stage 2 feed into our primary UAS design (detailed in the Concept Paper). The Concept Paper describes the specific airframe configuration, propulsion system, and structural approach we are proposing for the competition — informed by, but distinct from, the generic multirotor test platforms.

Why Test Multirotors Instead of the Final Design Directly?

Testing the final design first would conflate calibration errors with design errors. By validating physics on known, off-the-shelf platforms — where ground truth is well-characterized — we ensure our optimizer is working from accurate physical models. Only after that validation do we trust the optimizer's output as a guide for the competition design. This approach also lets us iterate quickly: multirotor test platforms are cheap and fast to reconfigure, while the primary design involves more specialized manufacturing.

Reference Documents

Document	Location	Description
Concept Paper	docs/Anote_Final_Concept_Paper.pdf	Full primary UAS design concept for DARPA LIFT
Physics model	src/darpalyft/core.py	Thrust, structural, and payload calculations
Optimizer	src/darpalyft/core.py — DesignOptimizer	Design search and feasibility filtering
Evaluation metrics	src/darpalyft/evaluate.py	Payload-per-weight scoring and Pareto analysis
Optimization pipeline	scripts/run_optimization.py	End-to-end run script

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